

ANNOUNCING PARAMETER

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- β is the Lagrange multiplier the rate at which the energy constraint trades off against entropy

$$S(\beta) := -k_B \text{KL}(\pi_\beta \| \eta)$$

$$= k_B \beta \langle U \rangle_\beta + k_B \log Z(\beta)$$

- We define the **temperature** T of a system as the reciprocal of β

$$T := \frac{1}{k_B \beta}$$

► β is often referred to as the inverse-temperature of the system

► Again, we can assume $k_B = 1$ but in the physics literature, you will see $\beta^{-1} = k_B T$

ANNEALING PARAMETER

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